

Clinical Trials IV Assignment 1

April 2026

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1 Introduction

This report outlines a randomised controlled trial for a treatment for Spontaneous Temporal Echo Episodes (STEE). The main symptom of STEE is perceiving events several minutes before they occur due to chronosynaptic misfiring, which can be disorienting and cause distress. It is known that it affects both sexes equally and is more common in people in their twenties, often resolving with age. It is believed that the frequency of echoes is impacted by stress. The population will consist of participants over the age of 16 who have experienced at least 3 echoes in the past 30 days. The primary outcome variable will be whether or not a participant experiences echoes within ten days of starting the trial. The protocol detailing allocation and analysis should be defined ahead of recruiting participants. We test the hypotheses:

$$\begin{aligned}\mathcal{H}_0 &: \pi_T = \pi_C, \\ \mathcal{H}_1 &: \pi_T \neq \pi_C,\end{aligned}$$

where π_T and π_C represent the probability of a participant experiencing an echo within ten days of starting the trial, given that they are in the treatment group and control group respectively. We note that $\pi_T < \pi_C$ corresponds to a positive outcome, since high probability of echoes is undesirable.

2 Sample Size

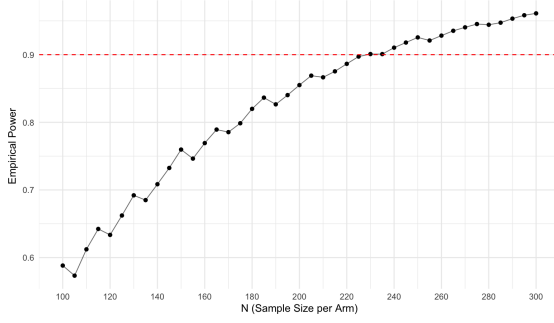
The choice of sample size is important to ensure sufficiently powered results of a trial. There exists a trade-off: recruiting too few participants results in insufficient power and an inconclusive trial, while too many participants may mean that they are given an inferior treatment unnecessarily when they could be receiving treatment. We would like to detect a reduction in the probability of experiencing an echo within ten days by at least 0.15, with a power of 0.9 at significance level 0.05. We choose our sample size for each trial arm, N , via simulation. We allocate the same number of participants to each arm as this maximises statistical power of the test, and allocation will ensure balance of characteristics between arms. The benefits of using simulation include transparency, since the data generating mechanism is clear and reproducible, flexibility, enabling us to explore the theoretical consequences of trial choices, and also the requirement to perform a ‘mock’ analysis before the trial, such that any issues can be rectified in advance. We simulate a two-sided chi-squared test for $N \in [100, 300]$ in increments of 5, estimating power as the proportion of 100,000 simulated trials that rejected $\mathcal{H}_0$ at the 5% level for each N . The results are shown in fig. 1a, with the desired power ($1 - \beta = 0.9$) given by the red dashed line, indicating that the test first crosses the power threshold for some $N \in [225, 230]$. Although fine-tuning indicated that $N = 227$ is the smallest value on N with sufficient power, this is subject to Monte Carlo uncertainty. An underpowered trial risks an inconclusive result, and so we will choose N slightly larger than this minimum value to leave a safety margin above the minimum value, accounting for randomness. We provisionally set $N = 235$, with the histogram of 200 independent trials under this value of N shown in fig. 1b, with a peak at power of 0.9, and estimates tightly concentrated within ± 0.005 of this value. To cross check this result, we apply the approximate sample size formula for binary outcomes, which is given by

$$N = \frac{(z_\beta + z_{\alpha/2})^2}{2 [\arcsin(\sqrt{\pi_T}) - \arcsin(\sqrt{\pi_C})]^2} \approx 231.15,$$

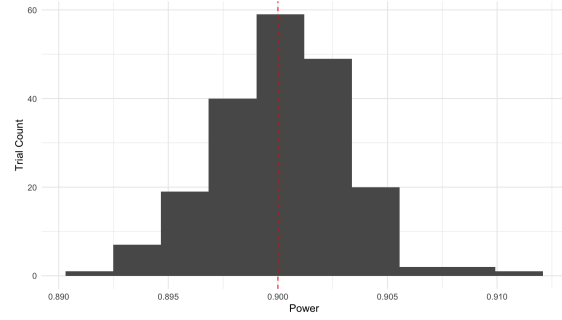
indicating that a minimum of 232 participants per arm are required, so that our chosen value of $N = 235$ sits above this lower bound. We note that this formula is an approximation based on the central limit theorem, but here it is expected to be accurate since π_C and π_T are not close to 0 or 1 and N is sufficiently large.

We adjust N to ensure the trial is robust against potential participant attrition which would reduce the power. Since the trial is fairly short, with the outcome assessed after 10 days of starting the trial, the attrition rate is anticipated to be relatively low. Based on an anticipated retention rate of 92% - the median retention rate for short trials (≤ 1 month) in a large-scale review of RCTs (Table 4, Jacques et al. 2022) - we calculate the new sample size per arm as:

$$N = \left\lceil \frac{235}{0.92} \right\rceil = 256.$$



(a) Empirical power calculated for different values of N averaged over 100,000 simulations with $\alpha = 0.05$, $\pi_C = 0.6$, $\pi_T = 0.45$. Desired power 0.9 (red dashed) first achieved for $N = 230$ (blue dotted).



(b) Histogram of empirical power for selected value of $N = 235$, with $\alpha = 0.05$, $\pi_C = 0.6$ and $\pi_T = 0.45$, centred on desired power of 0.9 (red dashed).

Figure 1

This results in a total sample size of 512 across both arms. We also note that $\pi_C = 0.6$ is stated as an estimate, and so sensitivity analysis would add depth, ensuring sample size is robust to small fluctuations in π_C .

3 Allocation

We now allocate the participants to the control and treatment arms. We measure four baseline covariates: age, sex, stress score (high, moderate or low) and the baseline STEE frequency, given by the number of episodes in the 30 days before allocation. It is known that STEE is equally common among males and females, most commonly affects those in their twenties, and tends to resolve with age. We therefore aim to balance all covariates in each arm. As a result, although simple random sampling is unbiased and is easy to implement, it is unsuitable for this trial as it gives no way to balance covariates, as well as potentially giving unequal trial arm sizes, which reduces power. Biased coin and urn methods have a similar disadvantage, as they also do not balance covariates, which can cause confounding if characteristics affect trial outcomes. Stratified random permuted blocks guarantees balance within strata, however having too many factors may result in small strata. Additionally, it requires binning of continuous variables, which is somewhat arbitrary and loses information. While minimisation also requires binning, it mitigates the small strata issue, as it can balance many covariates simultaneously. It works by calculating the hypothetical marginal imbalance with respect to each factor if the next participant were to be allocated the control group or the trial group. It introduces randomness allocating the current participant to the group which minimises the total imbalance with a high probability, for which we will use $p = 0.9$, so that there is a 0.1 chance the participant is allocated to the group which does not minimise imbalance, as this maintains some unpredictability. It can be argued that minimisation is ‘less random’ than the other specified methods, however this disadvantage is outweighed by the ability to successfully balance covariates, which is likely to be particularly important for STEE as we know that both age and stress level are thought to impact the frequency of echoes.

Both sex and stress score are already categorical variables, but age and baseline STEE frequency must both be binned. Since STEE is believed to be more common for people in their twenties and resolve with age, so we bin age into three categories: < 20 , $20 - 29$, and ≥ 30 . This captures those before and after the expected peak, as well as separating the twenties into two groups. Due to the trial recruiting only participants over 16, and the disease resolving with age, the < 20 and ≥ 30 may be quite small, so we monitor the balance in the allocation output. For the baseline STEE frequency, we split the data at the expected mean of 5, using two bins: < 5 and ≥ 5 . Participants are then allocated to arms sequentially via minimisation.

Characteristic		C	T
N		256	256
Age (Years)		24 (21, 28)	24 (22, 28)
Sex	F	121 (47%)	121 (47%)
	M	135 (53%)	135 (53%)
Stress Score	Low	123 (48%)	124 (48%)
	Moderate	86 (34%)	83 (32%)
	High	47 (18%)	49 (19%)
Baseline STEE Frequency		4 (3, 6)	4 (3, 6)

Table 1: Population characteristics by trial arm, presented as *count (proportion)* for factor variables sex and stress score, and *median (Q1, Q3)* for all continuous variables. Column C represents the control arm of the trial, and column T represents the treatment arm.

3.1 Allocation Results

The allocation rule was fixed in advance of recruitment to prevent selection bias and to allow strictly sequential allocation. We now examine post-allocation balance to verify the minimisation algorithm obtained balance of each characteristic across trial arms. In table 1, we see that the arms are of equal size and that we have successfully balanced the arms with respect to each of the covariates, with no meaningful imbalances. Specifically, we have a marginal imbalance of 2 for age and baseline STEE frequency, and a marginal imbalance of 6 for the stress score, in which the largest single-category imbalance is 3 (in the moderate stress group). The continuous variables show almost identical medians and interquartile range across trial arms.

4 Analysis

Since we used minimisation with respect to all baseline covariates for allocation, they must all be adjusted for in the analysis. We use the continuous data to ensure no loss of information.

4.1 Methodology

The below methodology is pre-specified before the outcomes are measured to prevent selection bias. We employ logistic regression, as is standard for binary outcome data, adjusting for all covariates, which is required since they were all used for minimisation. This results in the following model:

$$Y_i | \mathbf{x}_i \overset{\text{indep.}}{\sim} \text{Bernoulli}(p_i)$$

$$\log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_T G_i + \boldsymbol{\beta}^\top \mathbf{x}_i \quad (1)$$

where G_i is an indicator variable with value 1 if participant i was on the treatment arm and 0 otherwise, with coefficient β_T , and the other covariates represented by $\mathbf{x}_i$, with their coefficients $\boldsymbol{\beta}$, for each participant i , where $i = 1, \dots, 512$. Stress score and sex are factor variables, with reference levels ‘high’ and ‘female’ respectively.

We check for multicollinearity by calculating the Variance Inflation Factor (VIF), which reports a value close to 1 for each covariate, indicating no multicollinearity so that we can safely include all covariates in the analysis. No interaction terms are included a priori as there is no clinical reason to expect that the treatment effect of Chronquel differs depending on any of the baseline covariates. We will consider the odds ratio of the logistic regression model adjusted by covariates, as well as the absolute risk difference, number needed to treat, and risk ratio along with their associated confidence intervals. Wilson confidence intervals are used for the proportions because they guarantee that the confidence interval is inside $[0, 1]$, corresponding to natural bounds of proportions. They produce asymmetric intervals with better coverage. This makes Wilson confidence intervals a safer choice here, although they closely align with Wald confidence intervals

	Estimate	Std. Error	p-value	Odds Ratio	95% CI
Intercept	0.8300	0.7153	0.2459		
Age	0.02934	0.02014	0.14522	1.029	[0.9899, 1.071]
Sex - M	0.05367	0.1966	0.7848	1.055	[0.7178, 1.551]
Stress Score - Low	-1.692	0.3340	4.08×10^{-7}	0.1841	[0.09570, 0.3544]
Stress Score - Moderate	-1.051	0.3210	0.00106	0.3496	[0.1864, 0.6559]
Baseline STEE Count	0.05498	0.06647	0.4082	1.057	[0.9275, 1.204]
Arm	-1.351	0.1988	1.08×10^{-11}	0.2590	[0.1754, 0.3824]

Table 2: Result of applying the model (1) to the data. Significant p-values (red) for stress score (low and moderate) and trial arm.

for proportions not close to 0 or 1. We will then consider diagnostics including plotting a Receiver Operating Characteristics (ROC) curve and computing the area under the curve, as well as considering calibration.

4.2 Results

The results of the logistic regression are given in table 2, where treatment arm as well as low and moderate stress score are found to be significant. Their p-values (highlighted in red) indicate significance at the 5% level. The coefficient for arm (-1.351) corresponds to an odds ratio of approximately 0.26, indicating that the odds of an echo in the treatment arm are 26% of those in the control arm. The 95% confidence interval in terms of odds ratio is given by

$$\exp(-1.351 \pm 1.96 \cdot 0.1988) = [0.1754, 0.3824],$$

which suggests that the odds of an echo in the treatment arm are between 17.5% and 38.2% of those in the control, a reduction of between 61.8% and 82.5%. We note that this interval does not contain 1, and therefore we can be confident that the treatment reduces the odds of an echo occurring within the 10 day period. The odds ratios and confidence intervals for other covariates are given in the table, from which we note the insignificant variables have confidence intervals containing 1, as expected.

As well as the significance of the treatment effect, we note that low and moderate baseline stress scores are also significant in the occurrence of echoes, relative to high stress scores. Therefore those with high baseline stress scores had significantly higher probability of experiencing an echo, and so stress is an indicator for echo frequency regardless of treatment. Specifically, the odds ratio of approximately 0.18 for low stress compared to high means that low-stress participants had under one fifth of the odds of experiencing an echo of high-stress participants. This aligns with the prior belief that stress influences echo frequency, and verifies that balancing arms with respect to stress was important for unbiased results.

The age coefficient is positive but small and non-significant (OR 1.029, CI [0.9899, 1.071]), which may appear to contradict the claim that STEE improves with age. However, we note that the participants are required to have experienced at least 3 echoes in the past 30 days and so only those with active symptoms are considered. This also explains the narrow IQR of (21, 28).

We also consider the raw data, shown in table 3b. Running an unadjusted logistic model, we assume the data follows:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_T G_i,$$

where we have not adjusted for covariates. This results in a p-value of 6.6×10^{-11} , with an odds ratio of 0.30, and confidence interval [0.2081, 0.4296]. We see that considering the covariates strengthens the estimate of the treatment effect, since some of the variance is explained by the stress score. The fact that adjusted and unadjusted estimates yield similar results indicates the success of the minimisation in balancing the arms. To compute each of these, we use the raw data, shown in ??, which gives proportions $p_T = 90/256 = 0.3516$ and $p_C = 165/256 = 0.6445$, interpreted as the estimates for probability of experiencing an echo within ten

Covariate	VIF
Age	1.010
Sex	1.007
Stress Score	1.084
Baseline Stee Count	1.156
Arm	1.020

(a) VIF for each covariate

	Control	Treatment	Sum
0	91	166	257
1	165	90	255
Sum	256	256	512

(b) Treatment arm and count of observed 0s and 1s in the no-adjusted logistic regression model.

Table 3

days of receiving the treatment for each arm, where we note that p_T is significantly lower than p_C , indicating a positive treatment effect. We also note that the estimated control probability of 0.6445 is close to the assumed value of 0.6 used in the design stage, suggesting that assumptions were reasonable. We compute 95% Wilson confidence intervals for each of these, given by $[0.2957, 0.4119]$ for p_T and $[0.5842, 0.7006]$ for p_C indicating that more than half of the control group are estimated to experience echoes during the trial, compared to less than half of the treatment group.

Discrimination is a diagnostic checking how well the model is able to separate participants who experienced an echo within 10 days of the treatment from those who did not. The ROC curve (fig. 2) shows the false positive rate against the true positive rate, which sits above the $y = x$ line indicating it predicts the outcomes better than random guessing. It has an area under the curve (AUC) of 0.74, which is considered fairly strong, as it represents the model assigning higher probability to the correct arm 74% of the time. Another diagnostic to consider is calibration, which differs from discrimination as considers whether the predicted probabilities are accurate, whereas discrimination is only concerned with whether relative predictions are accurate, viewing the logistic model like a classifier. Calibration considers different combinations of baseline covariates and assesses whether the proportions predicted by the model match those of the observed data. An example using age (bucketed in 5 year intervals) and stress level is shown in fig. 3. These covariates were selected since age is known to affect instances of STEE and stress levels were found to be statistically significant in the regression, and baseline STEE was held at its median value. We see that the predicted probabilities align closely with the observed values, with the majority of large dots (representing more observations) within the 95% confidence interval of each predicted line, so that the model appears generally well-calibrated. In the case of moderate stress, many of the observations fall outside these bounds, however there is no systematic pattern of over or under-prediction so we attribute this deviation to randomness, rather than a mis-calibrated model.

We note that the treatment prediction line sits significantly lower than the control treatment line, highlighting the substantial treatment effect found in the logistic model. The gap between trial arms, which represents the ARD, is wider in the low and moderate stress plots. This is due to the non-linearity of the logistic function, which is steepest around 0.5, so the high stress difference appears smaller as the control probabilities (around 0.8) are further from the centre. All predictions show a slight upward trend as age increases, consistent with the small positive coefficient (0.029) in table 2.

Although the odds ratio is the natural output of logistic regression, it can be difficult to interpret. We therefore report multiple measures alongside the OR to provide clinically interpretable summaries. We consider the Absolute Risk Difference (ARD), Number Needed to Treat (NNT), and Risk Ratio (RR). The ARD is estimated by $p_C - p_T = 0.2930$ (negated from standard formula since $p_T < p_C$ is desirable in this case, with a Wilson confidence interval of $[0.2102, 0.3758]$). We can interpret this as a 29.3% reduction in the absolute probability of a participant experiencing an echo within 10 days. The trial was powered to detect a reduction of 0.15 in the probability of an echo within 10 days, whereas these results indicate a reduction of 0.293, which is almost double this minimum detectable effect.

Closely related to the ARD, the NNT is given by $1/\text{ARD}$, which is estimated as approximately 3.413 (CI $[2.661, 4.757]$) and interpreted as the number of patients you would need to treat before you would bring benefit to one extra patient. This means that treating an extra 4 patients, we would expect one of them to no longer experience the echo, which is a strong result. Finally, the risk ratio is estimated by $p_T/p_C = 0.5455$

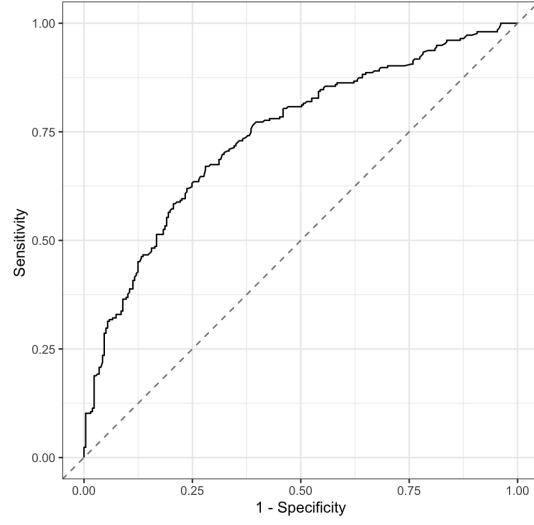


Figure 2: ROC curve for the adjusted covariate model, with AUC=0.74.

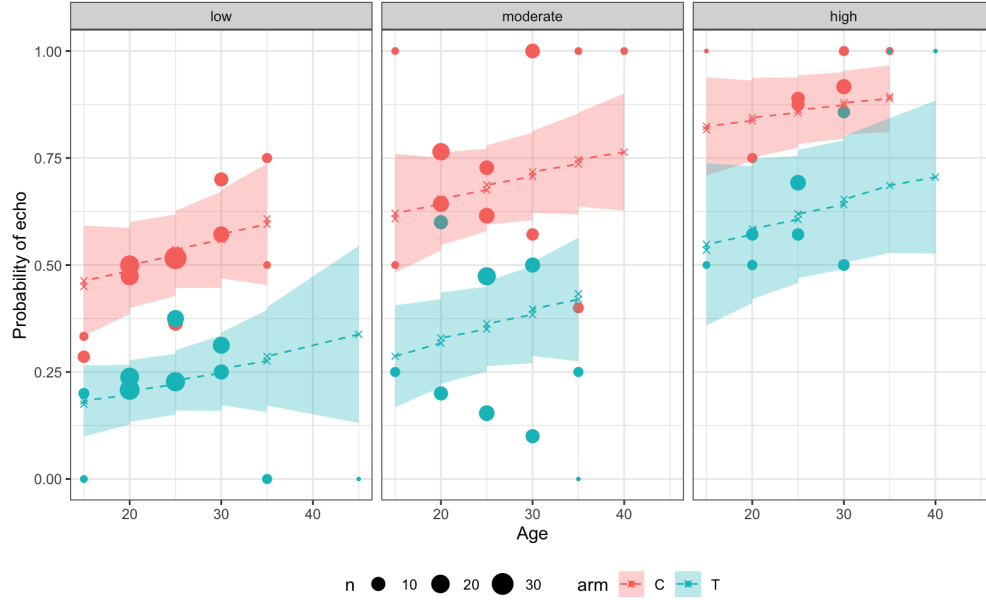


Figure 3: Calibration plot showing probability of an echo occurring for different values of stress level and age. Predicted values under the covariate adjusted model eq. (1) (dashed lines) with a 95% confidence interval shown for both the control (red) and treatment (blue) arms. True observations (circles) with number of observations fitting those criteria indicated by circle size.

(CI [0.4512, 0.6593], computed on the log scale), which means that the risk of an echo in the 10 day period in the treatment group is around 54.6% of that in the control group, so there is a 45.4% reduction in risk. All of these statistics are more intuitive to interpret than the odds ratio given above.

5 Trial Considerations

STEE is a rare condition, and we expect that it will take several months to recruit enough participants. Therefore, reducing the number of participants, while still maintaining the power of the trial, is beneficial here, so that if a treatment is effective it can be released more quickly to those suffering from the disease to alleviate their psychological distress.

One way in which the number of participants could be reduced is by switching to a continuous outcome variable as opposed to a binary one. Modelling the count of echoes experienced in the 10 day window, as opposed to whether or not there were any, is a natural alternative model. Each result then carries more information per participant than in the binary outcome case. Through the use of ANCOVA, correlation of baseline and outcome measurements, ρ , can reduce the variance by a factor of $1 - \rho^2$, and sample size scales linearly with variance. Since the baseline STEE count is measured, this can be used as a covariate in the regression model, exploiting its correlation with the outcome to reduce residual variance. However, since we expect 5 counts on average in 30 days, the results across 10 days may contain many zeros, and so ANCOVA may not be effective here since it requires that the counts are reasonably large and can be approximated by a normal distribution. One solution could be to extend the trial period, though this increases risk of attrition. Conversely, the binary outcome data followed a Bernoulli model with minimal assumptions. The use of linear regression introduces the assumptions of normal residuals and also homoscedasticity, neither of which are required in the binary outcome case. The interpretation of results would also differ, since the binary case gives a clear echo vs no echo distinction, whereas the continuous case requires more interpretation on what is classed as an improvement, although it could be informative to note whether the number of echoes decreased, rather than only interpreting zero echoes as a success. The analysis in this case would be based on residual plots, and diagnostics would involve quantile-quantile plots rather than ROC and AUC as above.

An alternative approach for sample size reduction would be use of a crossover method. This works by alternating the patients between the treatment and control groups, so that the between-subject variability is reduced, as the effects are compared to their own individual baselines. Two treatment groups exist, receiving the control and treatment in opposite orders to reduce temporal effects, systematic differences between the measurement periods which would otherwise lead to bias. This method reduces the variance by cancelling out the participant-specific baseline level, so that the precision depends only on the within-participant noise and not the between-participant noise. This reduction in variance therefore reduces the required sample size, and is applicable to STEE since the echoes are a recurring symptom. Potential issues with this approach include the effects of the treatment continuing during the control period so that there is no defined cut-off for the treatment impact. A solution to this is use of a washout period, which is a sufficient gap between the two treatments, but this requires knowledge of the treatment and its persistence, which may not be available. Additionally, it also results in a longer trial duration (more than double) if we continue to measure based on a 10 day window. This may discourage participants from participating and also increases the risk of attrition, which is a significant issue because both outcomes are required for effective analysis. The analysis would involve a two-sample t-test comparing within-subject differences across the two sequences, with the treatment effect estimated as half the difference of group means. This model could be implemented with either a binary or a count outcome, however a count outcome would reduce the sample size more significantly, for reasons discussed within the previous approach, reducing the sample size by two distinct mechanisms and therefore compounding their effects.

A third alternative is the use of Bayesian logistic regression, which treats the treatment effect as a random variable with a probability distribution. After specifying a prior representing pre-trial beliefs, the posterior can be updated based on observations. This posterior distribution allows direct inference such as credible intervals to quantify uncertainty in the estimate. The resulting confidence intervals are more interpretable

than in the frequentist framework, as they allow inference about the treatment effect directly, rather than the probability of the interval containing the true value. Rather than a binary reject/fail-to-reject of $\mathcal{H}_0$, the Bayesian approach can consider a range of equivalence, computing the probability of the treatment effect being within a superior region and probability of it being in an inferior region, giving a richer, more interpretable output than a single p-value. The number of participants can be reduced if there exists an informative prior based on previous data about the treatment. However, the choice of prior is inherently subjective and so a community of priors is required to be used, adding complexity to the analysis. The sample size can be computed by considering the expected posterior distribution, considering some criteria on the credible interval. A further advantage of this approach is the ability to update the posterior sequentially as data is observed. This means that the results can be monitored throughout the trial, which is not possible in frequentist analysis. Despite this, trial methodology should be outlined in advance of the trial, including decisions on early stopping. Moreover, the estimate $\pi_C = 0.6$ which is pre-specified in the frequentist case could be replaced by a prior representing uncertainty in π_C , rather than use of a fixed estimate, so that the power calculation is more robust.

6 Conclusion

The results of the primary analysis, covariate-adjusted logistic regression, indicated that the treatment Chronquel reduces probability of echoes in participants with STEE. The p-value was 1.08×10^{-11} and so the null hypothesis was strongly rejected. The odds ratio was 0.259, so that the odds were reduced to 26% of those of the control arm. We considered the simpler logistic regression without adjusting for covariates, from which we obtain sample proportions as probability estimates, with $p_C = 0.645$ and $p_T = 0.352$ for the control and treatment arms respectively. One particularly notable result is a 45.4% risk reduction of an echo in the treatment group when compared with the control. The trial was designed to power a reduction in probability of 0.15, and the actual reduction found was 0.293. The control group proportion was estimated to be 0.6 and was observed as 0.645. Both of these results are indicative of a well-calibrated trial. The close agreement in the adjusted (OR=0.26) and unadjusted (OR=0.30) treatment estimates confirms the success of the minimisation for allocation, producing arms with balance across each characteristic.

Despite the strong result, use of a binary outcome carries less information than a continuous outcome variable, for which number of echoes experienced is a natural choice. The baseline stress covariate was found to be significant, so that those with higher stress levels are likely to experience more echoes, with a 5 times higher number of echoes in those who identify as having high stress levels compared to those with low stress. The treatment was shown to be effective across all stress strata, as in fig. 3. The age range was fairly narrow, concentrated around roughly 18-30 years, likely as a result of the condition resolving with age. However, this could indicate that the findings may not generalise to older patients with STEE.

The results provide strong evidence that Chronquel reduces the probability of an echo within 10 days. Further trials could investigate the effects in longer periods, consider echo counts as outcome data, and potentially study patients over 30 in more depth. Furthermore, the significance of stress means that stress reduction techniques could be employed to mitigate symptoms for patients while awaiting release of a treatment and future trials should stratify by stress levels to maintain arm balance.

References

Jacques, Richard M et al. (2022). “Recruitment, consent and retention of participants in randomised controlled trials: a review of trials published in the National Institute for Health Research (NIHR) Journals Library (1997–2020)”. In: *BMJ Open* 12.2. ISSN: 2044-6055. DOI: 10.1136/bmjopen-2021-059230. eprint: <https://bmjopen.bmj.com/content/12/2/e059230.full.pdf>. URL: <https://bmjopen.bmj.com/content/12/2/e059230>.